

Potentials and Fields

Problem 10.1

Show that the differential equations for V and $\mathbf{A}$

$$\begin{aligned}\nabla^2 V + \frac{\partial}{\partial t}(\nabla \cdot \mathbf{A}) &= -\frac{1}{\epsilon_0} \rho \\ \left(\nabla^2 \mathbf{A} - \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{A}}{\partial t^2} \right) - \nabla \left(\nabla \cdot \mathbf{A} + \mu_0 \epsilon_0 \frac{\partial V}{\partial t} \right) &= -\mu_0 \mathbf{J}\end{aligned}$$

can be written in the more symmetrical form

$$\begin{aligned}\square^2 V + \frac{\partial L}{\partial t} &= -\frac{1}{\epsilon_0} \rho \\ \square^2 \mathbf{A} - \nabla L &= -\mu_0 \mathbf{J}\end{aligned}$$

where

$$\square^2 \equiv \nabla^2 - \mu_0 \epsilon_0 \frac{\partial^2}{\partial t^2} \quad \text{and} \quad L \equiv \nabla \cdot \mathbf{A} + \mu_0 \epsilon_0 \frac{\partial V}{\partial t}$$

Solution:

Put the definitions of $\square^2$ and L into the equations for V :

$$\begin{aligned}\left(\nabla^2 - \mu_0 \epsilon_0 \frac{\partial^2}{\partial t^2} \right) V + \frac{\partial}{\partial t} \left(\nabla \cdot \mathbf{A} + \mu_0 \epsilon_0 \frac{\partial V}{\partial t} \right) &= \\ = \nabla^2 V - \mu_0 \epsilon_0 \frac{\partial^2 V}{\partial t^2} + \frac{\partial(\nabla \cdot \mathbf{A})}{\partial t} + \mu_0 \epsilon_0 \frac{\partial^2 V}{\partial t^2} &= \\ = \nabla^2 V + \frac{\partial}{\partial t}(\nabla \cdot \mathbf{A}) = -\frac{1}{\epsilon_0} \rho \quad \# \end{aligned}$$

and for $\mathbf{A}$:

$$\begin{aligned} & \left(\nabla^2 - \mu_0 \epsilon_0 \frac{\partial^2}{\partial t^2} \right) \mathbf{A} - \nabla \left(\nabla \cdot \mathbf{A} + \mu_0 \epsilon_0 \frac{\partial V}{\partial t} \right) = \\ & = \left(\nabla^2 \mathbf{A} - \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{A}}{\partial t^2} \right) - \nabla \left(\nabla \cdot \mathbf{A} + \mu_0 \epsilon_0 \frac{\partial V}{\partial t} \right) = -\mu_0 \mathbf{J} \quad \# \end{aligned}$$

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